

Calculus I Lesson 26B

1) $f(x) = 3x^2 + 3x - 5, \quad x \in [-2, 3]$

Let $a = -2$ and $b = 3$

a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

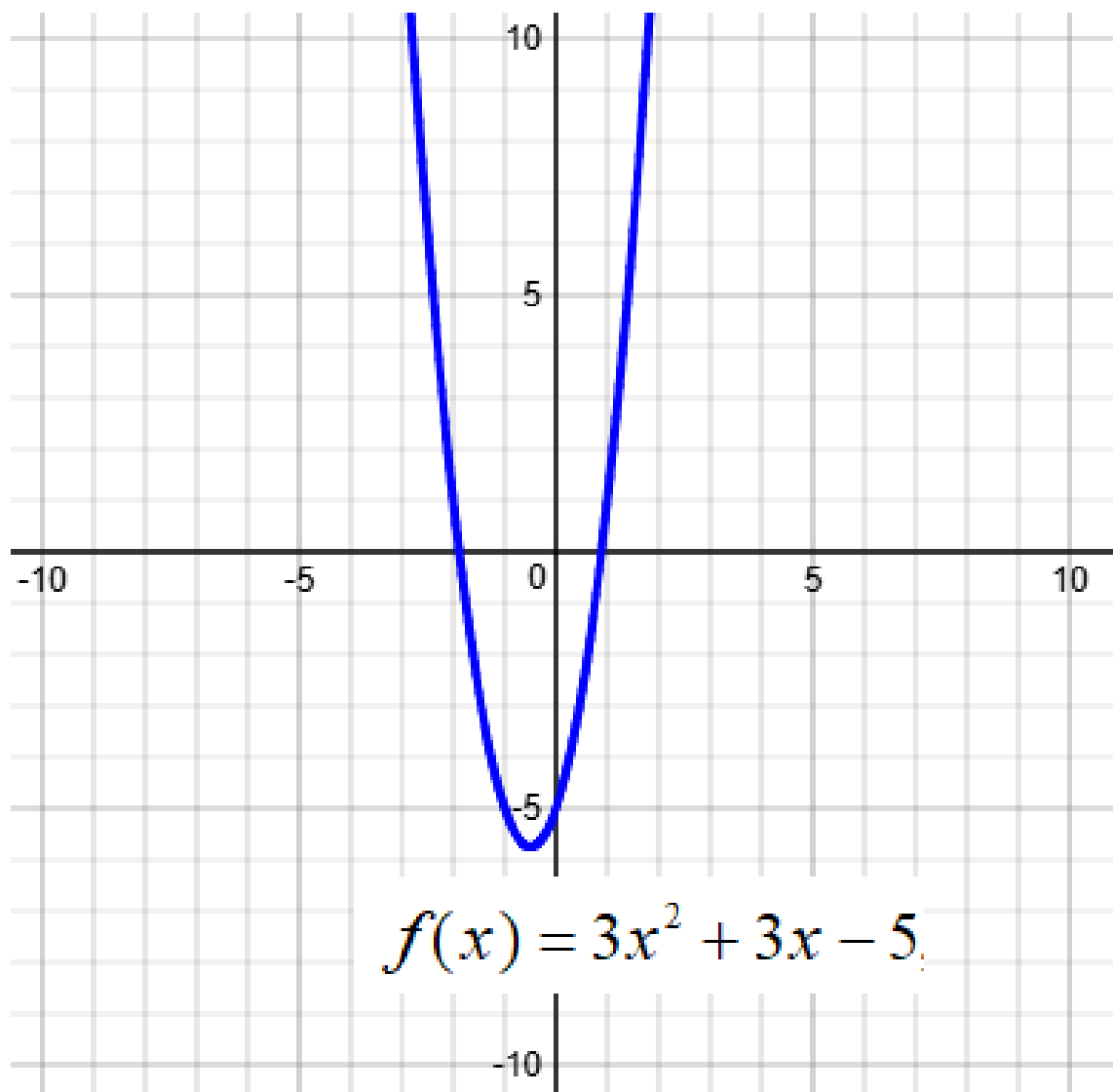
c) $\frac{f(b) - f(a)}{b - a} = \underline{\quad ? \quad}$

d) Mean Value Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , then there must be at least

one number $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Find c such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Hint: Find $f'(x)$, set $f'(x) = \frac{f(b) - f(a)}{b - a}$, and solve for x .



2) $f(x) = -x^2 + 8x - 5, \quad x \in [-1, 9]$

Let $a = -1$ and $b = 9$

a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

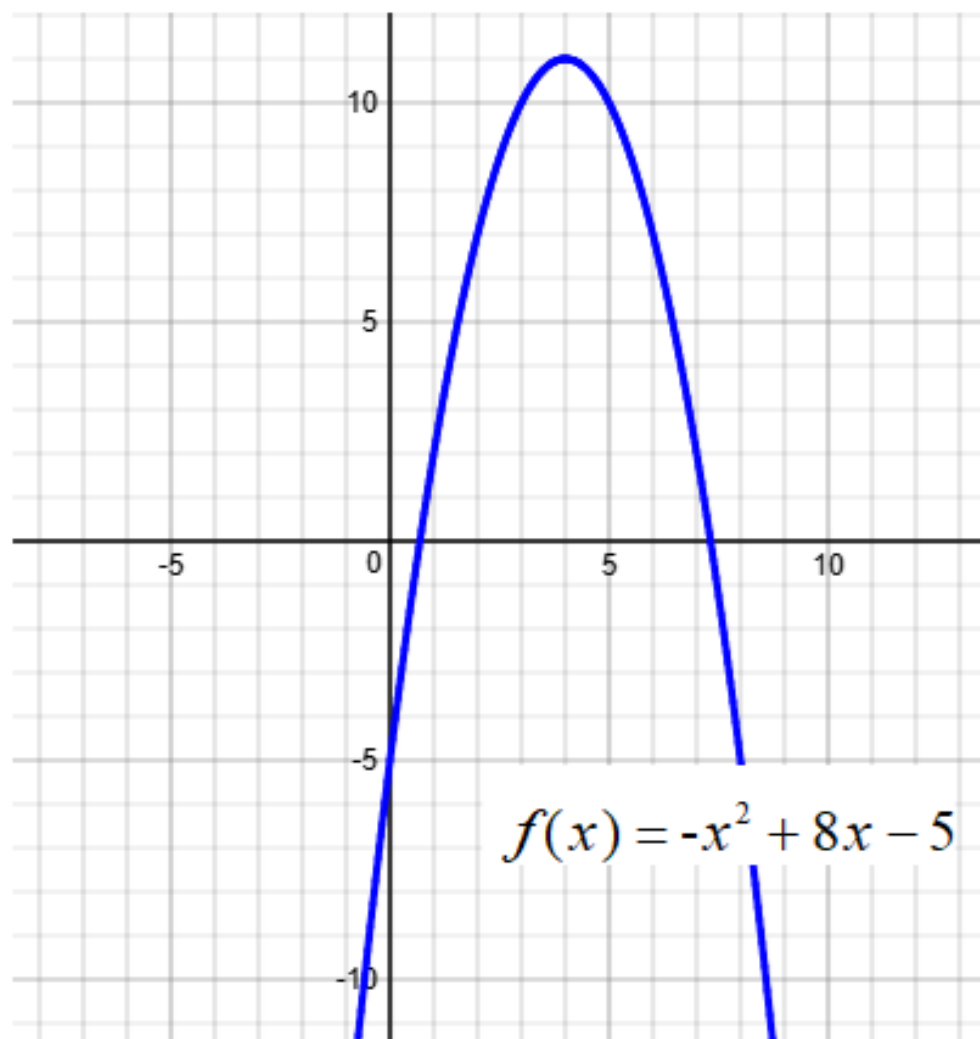
c) $\frac{f(b) - f(a)}{b - a} = \underline{\quad ? \quad}$

d) Mean Value Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , then there must be at least

one number $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Find c such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Hint: Find $f'(x)$, set $f'(x) = \frac{f(b) - f(a)}{b - a}$, and solve for x .



3) $f(x) = x^3 - 2x^2 - 5x + 6, \quad x \in [0, 4]$

Let $a = 0$ and $b = 4$

a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

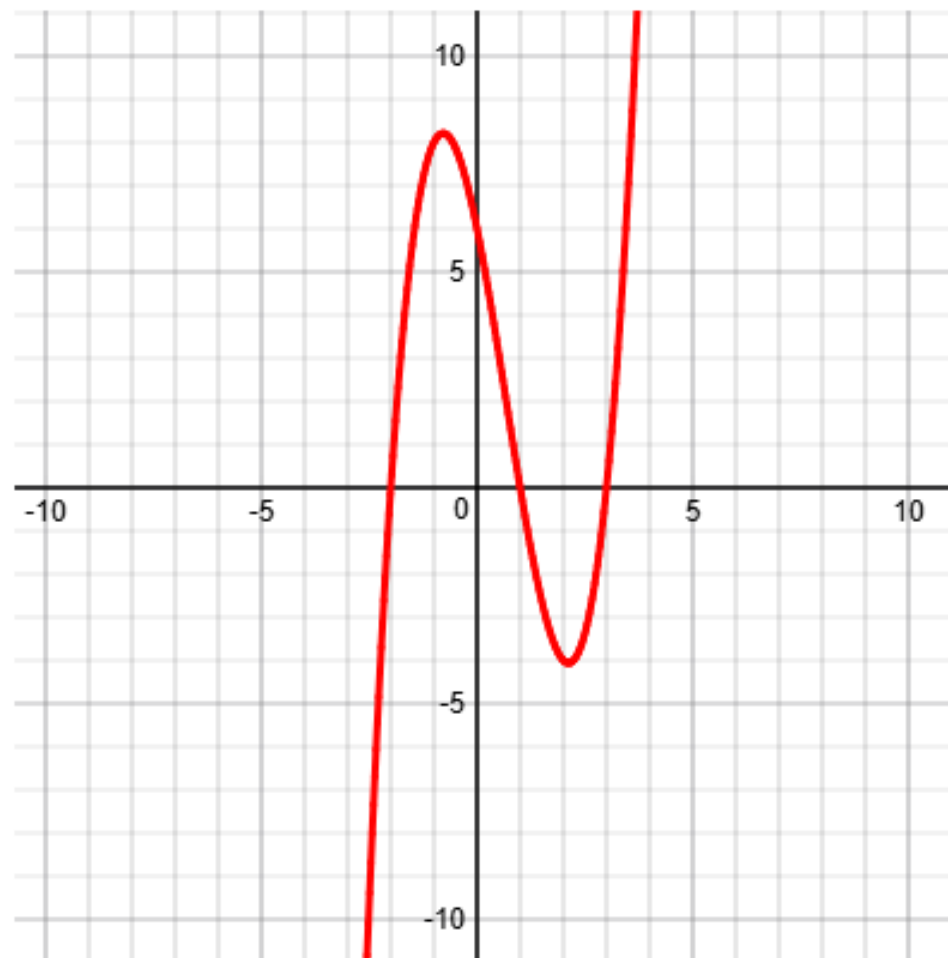
c) $\frac{f(b) - f(a)}{b - a} = \underline{\quad ? \quad}$

d) Mean Value Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , then there must be at least

one number $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Find c such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Hint: Find $f'(x)$, set $f'(x) = \frac{f(b) - f(a)}{b - a}$, and solve for x .



$$f(x) = x^3 - 2x^2 - 5x + 6$$

4) $f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$, $x \in [-1, 6]$

Let $a = 1$ and $b = 3$

a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

c) $\frac{f(b) - f(a)}{b - a} = \underline{\quad ? \quad}$

d) Mean Value Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , then there must be at least

one number $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Find c such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Hint: Find $f'(x)$, set $f'(x) = \frac{f(b) - f(a)}{b - a}$, and solve for x .

5) $f(x) = -0.5x^4 + 7.5x^2 + 5x - 12, \quad x \in [0, 2]$

Let $a = 0$ and $b = 2$

a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

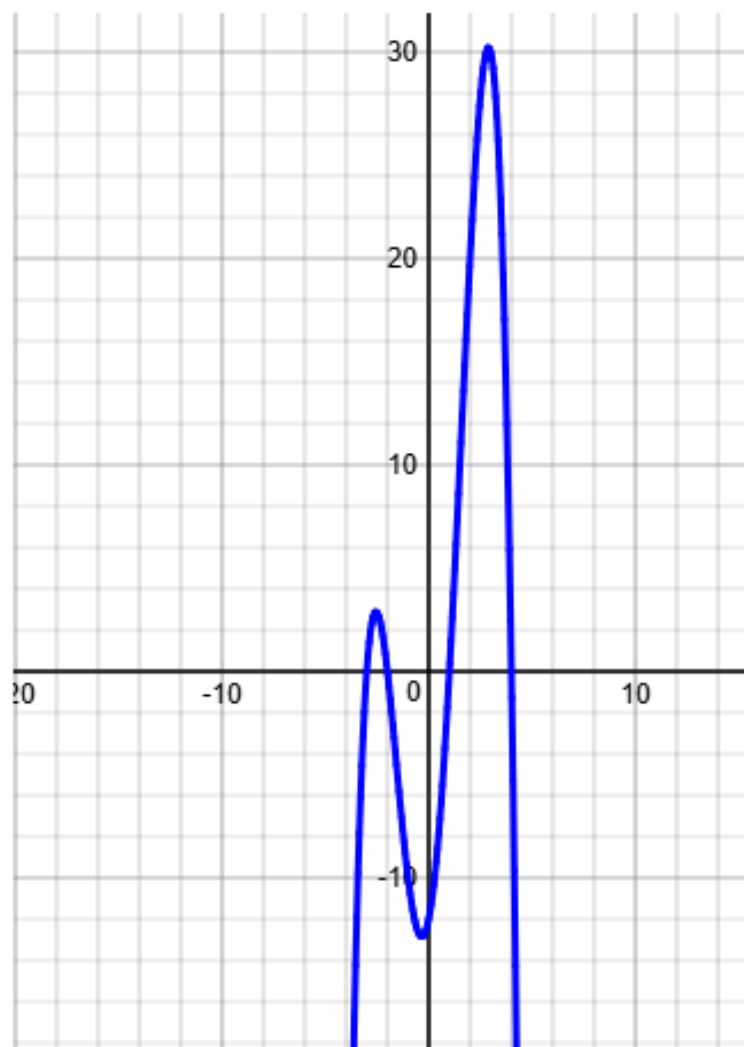
c) $\frac{f(b) - f(a)}{b - a} = \underline{\quad ? \quad}$

d) Mean Value Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , then there must be at least

one number $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Find c such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Hint: Find $f'(x)$, set $f'(x) = \frac{f(b) - f(a)}{b - a}$, and solve for x .



$$f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$$

6) $f(x) = 4 - \frac{3}{x}, \quad x \in [1, 5]$

Let $a = 1$ and $b = 5$

a) $f(a) = \underline{\quad ? \quad}$

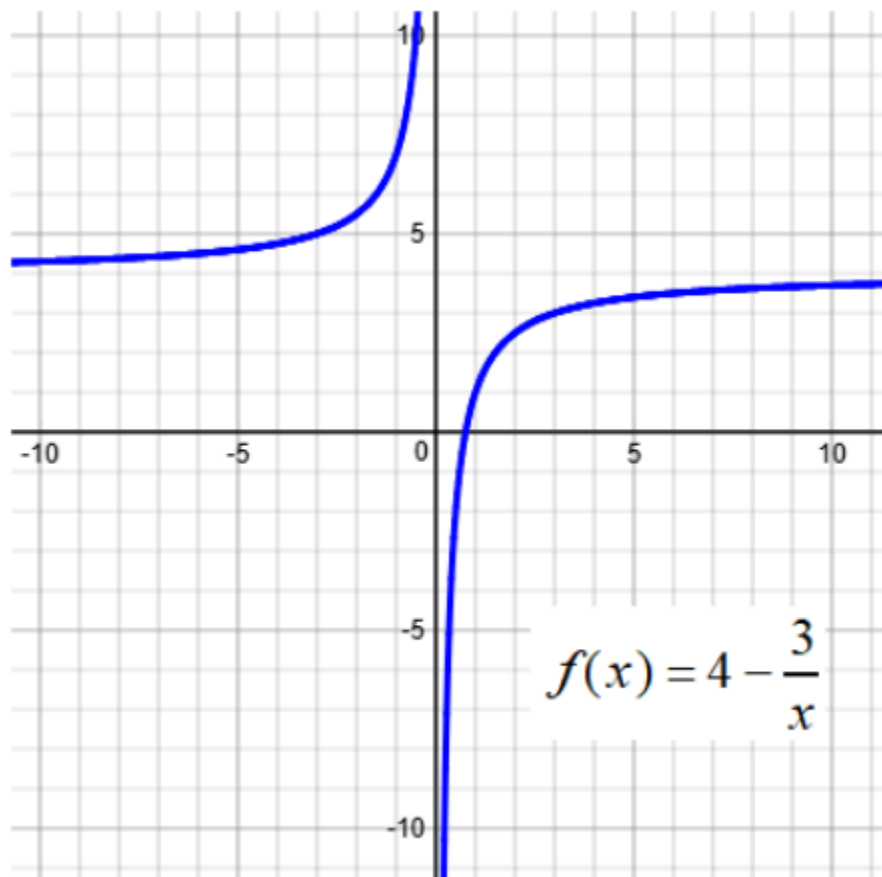
b) $f(b) = \underline{\quad ? \quad}$

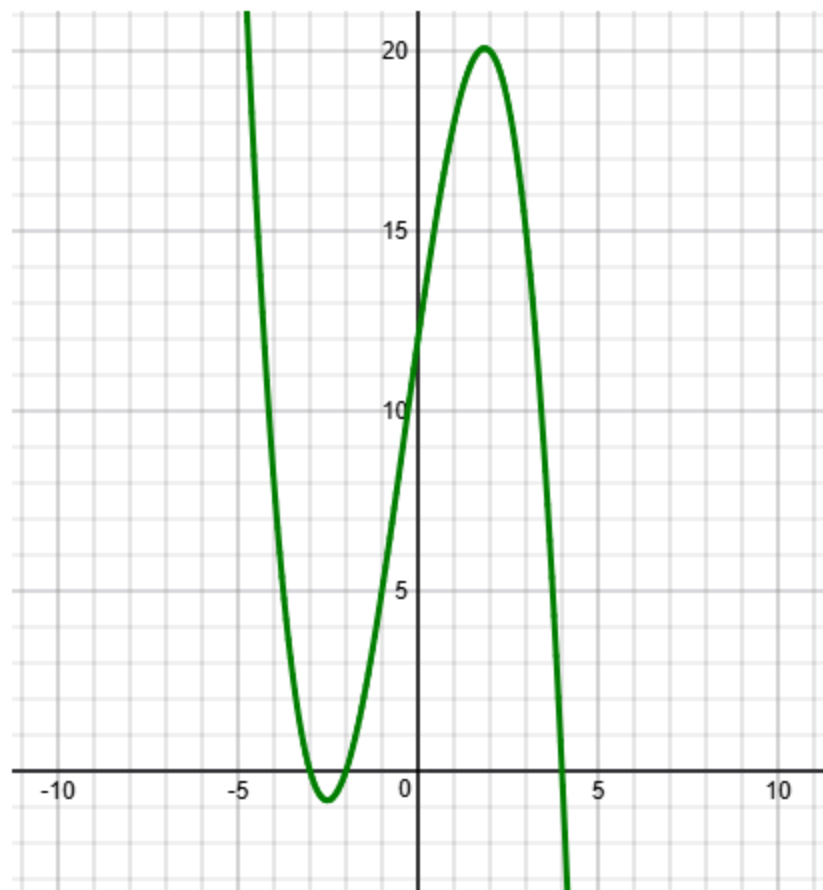
c) $\frac{f(b) - f(a)}{b - a} = \underline{\quad ? \quad}$

d) Mean Value Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , then there must be at least one number $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Find c such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Hint: Find $f'(x)$, set $f'(x) = \frac{f(b) - f(a)}{b - a}$, and solve for x .





$$f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$$

